



Tanvir Ahmed Fahim

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Summary

Electrical & Electronic Engineering graduate with strong technical, analytical, and problem-solving skills. Eager to apply my engineering knowledge and technical skills in a professional environment, contribute to organizational success, and build a strong career through continuous learning and development.

Education

2022 – 2026
Cumilla, Bangladesh

Bachelor of Science in Electrical & Electronic Engineering,
Bangladesh Army International University of Science & Technology (BAIUST)
CGPA: 3.81/4.00 (Thesis defense pending)

Professional Experience

02/09/2026 –
24/09/2026
Dhaka, Bangladesh

Industrial Trainee, Nagorik TV

- Gained practical exposure to television broadcasting operations, including studio production, control-room workflow, content management, and transmission systems.
- Developed an understanding of the end-to-end broadcast workflow involving Studio, PCR, MCR, Ingest, Editing, Newsroom, CER, Archive, and RF transmission

10/2020 – 05/2021
Cox's Bazar, Bangladesh

Wireless Operator, Field Intelligence Unit

- Operated secure wireless communication systems to support field intelligence and operational activities.
- Maintained reliable communication between teams while following communication protocols, security procedures, and operational instructions.

07/2019 – 04/2022
Bangladesh

Soldier, Bangladesh Army

- Served in multiple operational roles within the Bangladesh Army, demonstrating strong discipline, teamwork, and responsibility.
- Developed the ability to work effectively under pressure while maintaining professionalism and operational readiness.

Skills

Technical Skills: Microsoft 365, CST Studio Suite, HFSS, COMSOL, ADS, LTSpice, PSpice, Proteus, PSCAD, MATLAB, Python, C, Arduino, AutoCAD, SolidWorks, Simulink, Origin, SigmaPlot, Visio.

Professional Skills: Teamwork & Collaboration, Leadership, Communication, Problem-Solving, Time Management, Adaptability, Presentation Skills, Technical Writing, Ability to Work Under Pressure.

Projects

07/2026

ESP32-Based Multimodal Health Monitoring System

- Designed and developed an ESP32-based multimodal health monitoring system integrating a MAX30100 pulse-oximeter, three-electrode EEG sensor, and OLED display for real-time physiological-signal acquisition, with Wi-Fi-enabled remote monitoring through the Blynk IoT platform

12/2024

Smart Car Parking System

- Developed an IoT-based smart parking system utilizing IR obstacle sensors, servo motors, and LED indicators for automated gate control and real-time occupancy monitoring, reducing manual effort and improving efficiency.

08/2024

Design and Implementation of FIR Filters for Image Processing Applications

- Implemented FIR filter design techniques to enhance edge detection and reduce noise in image processing applications, improving accuracy in medical imaging diagnostics and security surveillance systems.

Languages

Bangla

Native language; excellent proficiency in speaking, reading, and writing.

English

Proficient in speaking, reading, writing, and professional communication.

Hindi

Good conversational proficiency in speaking and understanding Hindi.

Publications

- 30/08/2026 **Machine learning-driven gain prediction of a high-performance dual-wideband decoupling-enhanced graphene MIMO antenna for THz applications** , *Diamond & Related Materials* (Elsevier)
- 03/12/2025 **28 GHz Patch Antennas for 5G Wireless Applications: A Comprehensive Review** , *International Conference on Edge Computing and Applications* (Springer Nature)
- 16/04/2026 **Design of a High Performance Nano Band Microstrip Antenna with Three Rectangular Side Slots for THz Applications** , 2026 IEEE 2nd International Conference on Quantum Photonics, Artificial Intelligence & Networking (IEEE)
- 29/11/2024 **Enhancing Power Grid Reliability in Bangladesh: An IoT-Based Approach to Transmission Line Fault Detection** , 2024 IEEE 3rd International Conference on Robotics, Automation, Artificial-Intelligence and Internet-of-Things (IEEE)

Community Service & Volunteering Activities

- 11/2024 – 10/2025 **IEEE BAIUST Student Branch, Senior Executive Member**
 - Contributed to student branch activities and supported the planning and coordination of academic and professional initiatives.
 - Collaborated with fellow members to promote student engagement and organizational activities.
- 09/2024 – 10/2025 **BAIUST Robotics Society, General Secretary**
 - Led and coordinated robotics society activities, technical events, and student engagement initiatives.
 - Worked with team members to organize programs, manage responsibilities, and promote robotics and technology-based learning.
- 2017 – 2018 **Bangladesh National Cadet Corps (BNCC), FGC Platoon, Cadet**
 - Participated in cadet training and organizational activities, developing discipline, leadership, teamwork, and a strong sense of responsibility.
- 2014 – 2016 **Shaheen Academy School and College Scout Group, Scout Member**
 - Participated in scouting activities, parade drills, community service, and team-based programs, developing discipline, leadership, teamwork, and social responsibility.

Achievements & Honors

- 08/12/2025 **Best Poster Award**, BAIUST Inter-Department Poster Presentation Competition
- 2022 – 2024 **VC List Merit Tuition Waiver (10%–40%)**, BAIUST
Awarded for academic excellence in the 1st–4th and 6th–7th semesters.
- 12/2021 **Best Soldier Award 2021**, 9 Signals Battalion, Bangladesh Army
- 02/2016 **Scholarship on Board of Secondary School Certificate**

References

Dr. Mohammad Shafiul Alam, Associate Professor & Head,
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